Embedding Generalized Semantic Knowledge Into Few-Shot Remote Sensing Segmentation

Qi Wang, *Senior Member, IEEE*, Yuyu Jia, Wei Huang, Junyu Gao, *Member, IEEE*, and Qiang Li, *Member, IEEE*

***Abstract*— Few-shot segmentation (FSS) for remote sensing (RS) imagery leverages supporting information from limited annotated samples to achieve query segmentation of novel classes. Previous efforts are dedicated to mining segmentation-guiding visual cues from a constrained set of support samples. However, they still struggle to address the pronounced intra-class differences in RS images, as sparse visual cues make it challenging to establish robust class-specific representations. In this article, we propose a holistic semantic embedding (HSE) approach that effectively harnesses general semantic knowledge, i.e., class description (CD) embeddings. Instead of the naive combination of CD embeddings and visual features for segmentation decoding, we investigate embedding the general semantic knowledge during the feature extraction stage. Specifically, in HSE, a spatial dense interaction (SDI) module allows the interaction of visual support features with CD embeddings along the spatial dimension via self-attention. Furthermore, a global content modulation (GCM) module efficiently augments the global information of the target category in both support and query features, thanks to the transformative fusion of visual features and CD embeddings. These two components holistically synergize CD embeddings and visual cues, constructing a robust class-specific representation. Through extensive experiments on the standard FSS benchmark, the proposed HSE approach demonstrates superior performance compared to peer work, setting a new state-of-the-art.**

***Index Terms*— Class description (CD) embeddings, few-shot segmentation (FSS), remote sensing (RS), semantic embedding.**

I. Introduction

SEMANTIC segmentation, entailing the pixel-level categorization of images, is essential for deciphering remote sensing (RS) imagery and acts as a foundational method across a wide range of practical applications [1], [2], [3], [4], [5], [6]. With the development of aerial and satellite imaging devices, the collection of RS data becomes feasible, further driving advancements in deep learning for fully supervised RS image segmentation. However, deep learning-based methods [7], [8] still struggle with the high cost of extensive annotation data. Although researchers mitigate the stringent requirement for annotated data through semi-supervised [9], [10], [11], [12] and weakly supervised [13], [14], [15], [16] perspectives, the generalization ability of models remains a bottleneck when confronted with entirely new categories. As an effective approach to surmount these challenges, a plethora of few-shot segmentation (FSS) techniques [17], [18], [19], [20], [21], [22] have emerged. Given a few annotated samples from a novel class, i.e., *support*, FSS performs accurate segmentation on other samples within the same category, i.e., *query*.

As Fig. 1(a) shows, FSS algorithms typically follow the segmentation-guided paradigm, exploring the extraction of class-specific visual cues from support samples and segmenting query samples through a decoder [24]. Some efforts are devoted to the segmentation decoding stage. Specifically, prototypical learning methods [25], [26], [27] employ class prototypes as stand-ins for pixel-level correlations, indirectly guiding the segmentation of query images. Affinity learning methods [28], [29], [30] directly transform pixel-level correlations into segmentation guidance information. Another school of thought mines robust and comprehensive class-specific visual representations during the feature extraction stage, serving as segmentation-guiding cues. For example, integrating multiscale features [31], [32], [33] leveraging deep semantic correlations [34], [35], [36] and incorporating multi-level support class information [37], [38], [39], etc. Although showing promising results, these methods based on a single visual modality still have limitations. A minuscule number of samples often falls short in furnishing robust visual representations for segmentation guidance, especially when dealing with RS images characterized by pronounced intraclass differences, as illustrated in Fig. 2.

To address the above issue, introducing general semantic knowledge, i.e., class description (CD) embeddings, can supplement some missing class information in visual representations since CD embeddings are responsible only for categories and are not influenced by individual sample differences. Following this idea, AM3 [40] extracts textual prototypes from category descriptions and combines them with visual prototypes. SP [41] modulates visual feature extraction using semantic information as prompts. However, these methods are designed for classification tasks. Recently, MIANet [23] simply supplements CD embeddings with visual representation and achieves remarkable results [Fig. 1(b)], which indicates its promising development space.

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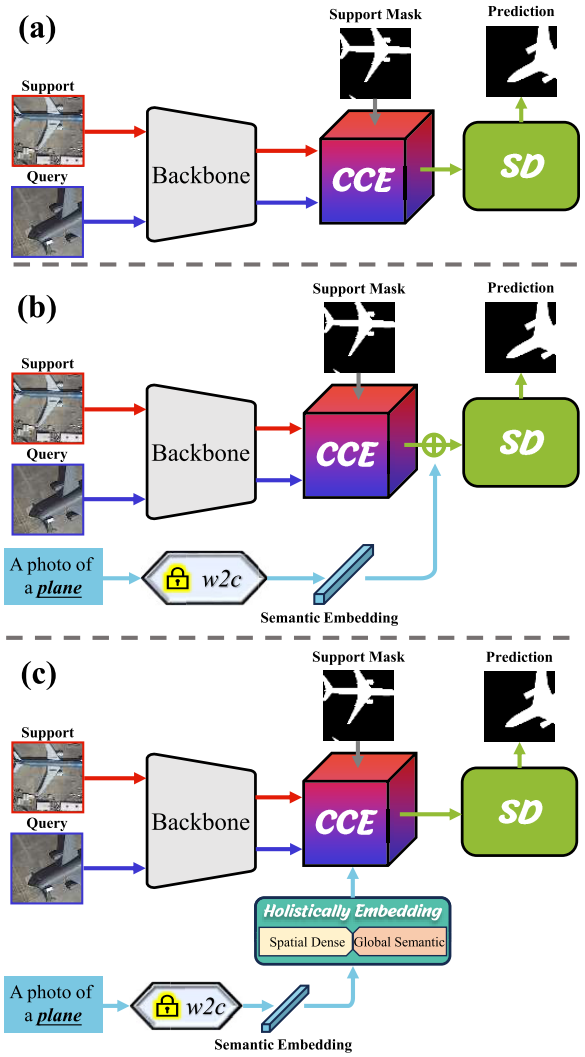


Fig. 1. Comparison between existing FSS methods and proposed HSE. (a) Many FSS algorithms adhere to a segmentation-guiding paradigm, primarily conducting research from two aspects: class-specific cues extraction (CCE) and SD. (b) Recent work, MIANet [23], introduces general semantic knowledge and combines it with visual support prototypes for segmentation guidance. (c) We further explore a HSE approach that exploits the capabilities of general semantic knowledge through SDI and global semantic modulation.

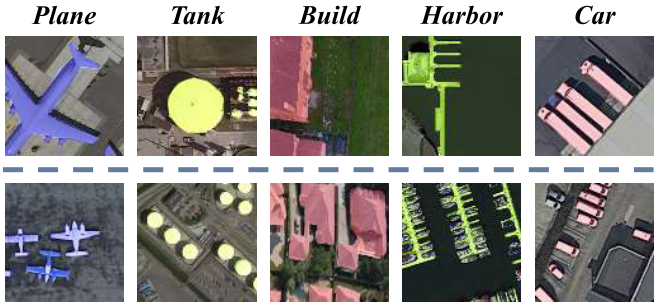


Fig. 2. Pronounced intraclass differences in RS images. Even though the same object category is present in the first and second rows, their appearance, size, and distribution characteristics differ remarkably.

In this article, we further explore a holistic semantic embedding (HSE) approach to harness the potential of general semantic knowledge. First, CD embeddings for each class can be obtained by a powerful pretrained language model, such as BERT [42] and CLIP [43]. Multilevel visual features are derived through the CNN backbone, i.e., ResNet50 [44] or VGG16 [45]. Subsequently, HSE is implemented as two serial complementary modules, i.e., spatial dense interaction (SDI) and global content modulation (GCM), to holistically embed general semantic knowledge into the visual feature extractor [Fig. 1(c)]. In particular, SDI extends the support features with general CD embeddings to facilitate the dense interaction *within the spatial dimension*. These enriched features are then processed through an interactor, which is realized using a self-attention mechanism, to guide the model to attend to spatial characteristics unique to each class. Moreover, the global content matters for comprehensively understanding the category scene. GCM converts CD embeddings and visual support prototypes into an enhancement coefficient through a modulator to enhance the support and query features *on the channel dimension*. Combining the two complementary modules, the proposed HSE approach effectively leverages the general semantic knowledge in CDs to mitigate intraclass individual differences in RS imagery. Through extensive experiments on the standard FSS benchmarks, i.e., iSAID-5^i, HSE presents significant performance improvements with different types of pretrained language encoders and few-shot settings, realizing the efficacy of general semantic knowledge for FSS tasks.

In summary, our contributions are threefold are listed below.

1) A novel HSE approach is proposed to explore general semantic knowledge in CDs. To our best knowledge, we are the first to introduce textual modality information in few-shot RS segmentation.
2) To establish robust class-specific representations for segmentation guidance, we propose two complementary modules that holistically embed semantic knowledge into the visual feature extractor from both spatial dense and global content perspectives.
3) Through comprehensive quantitative and qualitative analysis, we demonstrate the effectiveness and state-of-the-art performance of HSE on the FSS benchmark.

II. Related Works

A. Generic Semantic Segmentation

Semantic segmentation can provide pixel-level understanding for RS images and has been extensively studied in recent years, driven by advancements in deep neural networks. The introduction of fully convolutional networks (FCNs) [7] has pioneered the development of CNN-based methods. Following this foundation, U-Net [46] introduces an encoder–decoder structure with skip connections, enriching the feature space information. To enhance the model's multiscale perception capability, a series of variants of atrous convolution [8], [47], [48] and adaptive pooling [49], [50] techniques have emerged. Later, transformer-based methods [51], [52], [53] aim to balance local and global spatial features, effectively improving semantic segmentation performance.

Drawing from successes in natural scenes, some researchers are dedicated to the task of RS semantic segmentation. Considering the broader scale variations of RS objects, advanced

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methods predominantly employ attention mechanisms to extract global contextual information. ResU-Net [54] designs a linear attention mechanism to approximate dot-product attention with significantly reduced computation costs, rendering the integration of attention mechanisms and CNN more flexible. Given this, MANet [55] integrates global dependencies in both channel and spatial dimensions. SLCNet [56] supervises long-range correlations through category consistency information in the ground truth segmentation map. CGGLNet [57] leverages global contextual information to address unclear boundaries and incomplete structures. However, the aforementioned methods are limited by the requirement for large-scale data training and struggle to adapt to new scenes.

B. Few-Shot Learning

Few-shot learning endeavors to discern new categories with a limited set of labeled examples. Recent investigations reveal two quintessential directions: Optimization-based methods [58], [59], [60], [61] learn a meta-learner to perform rapid adaption using a sparse set of training samples for novel categories. Metric-based methodologies [62], [63], [64], [65] cultivate a feature space wherein a suitable distance function is applied for the assessment of similarity. More recently, large-scale language models (LLMs) and vision-language models (VLMs) have propelled advancements in few-shot learning. Some superior methods ingeniously leverage textual modal knowledge to compensate for the absence of visual features [41] or to directly enhance the few-shot classifiers [40].

In the realm of RS image processing, few-shot learning has likewise achieved significant progress. For example, RS-MetaNet [66] elevates the learning paradigm from individual samples to tasks through meta-training, thereby acquiring the ability to discern a metric space adept at classifying RS scenes. DLA-MatchNet [67] employs the attention mechanism to discover discriminative regions. HSL-MINet [68] proposes a multiview-attention strategy designed to extract potentially shared information across various rotational perspectives of images. MVP [69] incorporates a parameter-efficient tuning method into the meta-learning framework, customizing it specifically for RS images. ICSFF [70] amplifies feature representations for few-shot classification through a graph-structural feature fusion methodology. Although these methods have achieved notable success in classification tasks, they lack the capability for pixel-level understanding of RS images in few-shot scenarios.

C. Few-Shot Segmentation

Few-shot semantic segmentation aims to decode objects belonging to previously unseen categories with merely a handful of annotated samples. OSLSM [17], as a pioneering work in FSS, introduced a dual-branch segmentation paradigm. Following this, various approaches can generally be categorized into two groups according to their segmentation decoding strategies: prototypical learning methods [22], [25], [71] and affinity learning methods [28], [30], [72].

In the latest research, a state-of-the-art endeavor, MIANet [23] leverages the general semantic knowledge in category descriptions to enhance the representation of support samples, achieving promising results. However, inherent differences between RS and natural images—such as pronounced intraclass variations, larger scene extents, and richer elements—impede MIANet's effective performance in the FSS of RS images.

Similarly, excellent research in few-shot RS segmentation has increasingly emerged. For instance, the widely popular dataset iSAID-5^i is introduced in [73]. DMML-Net [74] constructs a deep feature pyramid comparison module and conceptualizes segmentation as a metric-based pixel classification task. PCNet [75] progressively parses the entirety of the target region into local descriptors and designs base-class memories to distillate prototypes for novel classes. HMRE [76] enhances the target information within the dual-branch network structure through global semantic and spatially dense mutual reinforcement. R2Net [77] implements dynamic global prototypes to mitigate inaccurate activations arising from intraclass differences. MS2A2Net [78] proposes an attention aggregation network to adaptively fuse multiscale features and effectively model the foreground correlation of the dual-branch structure. However, these methods are limited to extracting visual segmentation cues from a finite number of support samples and struggle with the significant intraclass differences in RS images. This article pioneers the incorporation of textual modal information from category descriptions in this task, addressing the shortfall of support information and providing fresh insights for this research.

III. Methodology

A. Task Description

FSS seeks to generalize a model's segmentation capability from a training dataset $\mathcal{D}_{\text{train}}$ to a completely novel category dataset $\mathcal{D}_{\text{novel}}$, corresponding to two disjoint sets of categories, $\mathcal{C}_{\text{train}}$ and $\mathcal{C}_{\text{novel}}$. Consistent with prior studies [26], we train the model episodically, where each episode includes a support set $\mathcal{S}$ and a query set $\mathcal{Q}$. Given the K-shot setting, each support set comprises K pairs of images I_s and their corresponding binary masks M_s, where $\mathcal{S} = \{I_s^k, M_s^k\}_{k=1}^K$. Similarly, the query set is defined as $\mathcal{Q} = \{I_q, M_q\}$, where the mask M_q of the query image is only available at the model training stage. Our goal is to mine robust segmentation-guiding information from the support set to accurately parse the query images. Note that we employ the one-shot setting to simplify the illustration of our approach.

B. Method Overview

As depicted in Fig. 3, the proposed HSE approach encompasses an initial feature extraction stage Section III-C and two key complementary modules, the SDI module Section III-D and the GCM module Section III-E, that explore general semantic knowledge from category descriptions, constructing robust class-specific representations for segmentation guidance. Specifically, given the support image I_s and the query image I_q, two pretrained backbones (ResNet50 [44] or VGG16 [45]) with shared weights are adopted to extract mid- and high-level features. We then obtain the CD embedding, support

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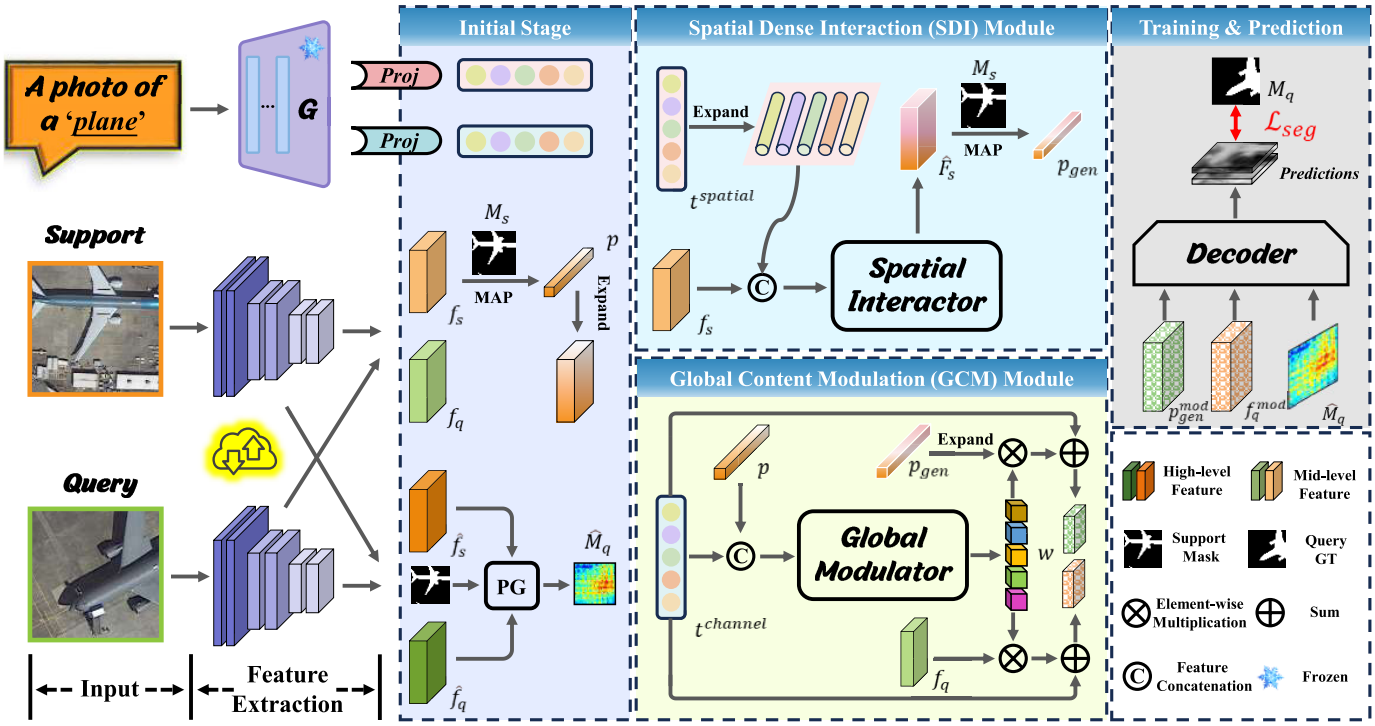


Fig. 3. Pipeline of the HSE method: it first extracts mid- and high-level support and query features, query prior masks, and CD embeddings in the initial feature extraction stage. To embed the general semantic knowledge from CD embeddings into visual cues and establish robust class-specific segmentation guidance, we design two sequential, complementary modules. The SDI module facilitates the SDI between general semantic knowledge and individual-specific visual features. The GCM module enhances global content relevant to the target category in the support and query features through modulation coefficients. Finally, along with the modulated query feature and query prior mask, the constructed robust class-specific representation assumes the role of segmentation guidance inputted into the decoder, yielding the query prediction mask.

prototype, and query prior mask in the initial feature extraction stage. Subsequently, the holistic embedding of general semantic knowledge is executed in two sequential modules, i.e., SDI and GCM. At last, we feed the robust class-specific guidance, the modulated query feature, and the query prior mask into a segmentation decoder (SD), obtaining the prediction query mask.

C. Initial Feature Extraction Stage

The core issue in FSS is how to transform the provided support image and its mask into class-specific segmentation guidance information. Among a plethora of studies, the predominant strategy is to compress the foreground region features of support images into prototypes, which are then used to guide the segmentation process of query images. To elaborate, background information in the middle-level support features is masked out, leaving only foreground information, which is then concentrated in the prototypes through average pooling. Formally, given the mid-level support feature $f_s \in \mathbb{R}^{C\times H\times W}$ and the corresponding support mask $M_s \in \mathbb{R}^{H\times W}$, where C, H, W represent the channel, height, and weight, the prototype $p \in \mathbb{R}^{C\times 1\times 1}$ can be expressed as follows:

$$p = \mathcal{A}(f_s \otimes \mathcal{R}(M_s)) \tag{1}$$

where $\mathcal{A}(\cdot)$ represents the average pooling operation, $\otimes$ denotes the Hadamard product, and $\mathcal{R}$ is a spatial scaling function.

Regarding the utilization of high-level features, the esteemed PFENet [26] transforms them into a class-agnostic prior mask to roughly indicate the target area's location in the query image. The prior mask $\hat{M}_q \in \mathbb{R}^{H\times W}$ is derived from the maximal element-wise correlation responses between the high-level features of the query and support. We abstract this process as a prior mask generator $\mathcal{PG}$, which takes support masks M_s, high-level features $\hat{f}_s$, and $\hat{f}_q$ as inputs. This can be represented as follows:

$$\hat{M}_q = \mathcal{PG}(\hat{f}_s, \hat{f}_q, M_s). \tag{2}$$

Moreover, with the pretrained language model $\mathcal{G}(\cdot)$, we transform the category *name* into the CD embedding $t = \mathcal{G}(\text{name}) \in \mathbb{R}^{C_t\times 1\times 1}$, where C_t is the dimension of the embedding space.

D. Spatial Dense Interaction

Existing studies revolve around the two-level segmentation guidance cues, i.e., p and $\hat{M}_q$, which are finally passed to the decoder along with the mid-level query features for segmenting the target region. However, they still struggle to break through the limitations imposed by the single visual modality, especially in the face of RS images that exhibit particularly substantial intraclass differences. Essentially, the segmentation guidance cues p and $\hat{M}_q$ are both extracted from sparse, individual-specific visual features. They lack sufficient general category information to construct robust class-specific representations, rendering it arduous to efficiently encode

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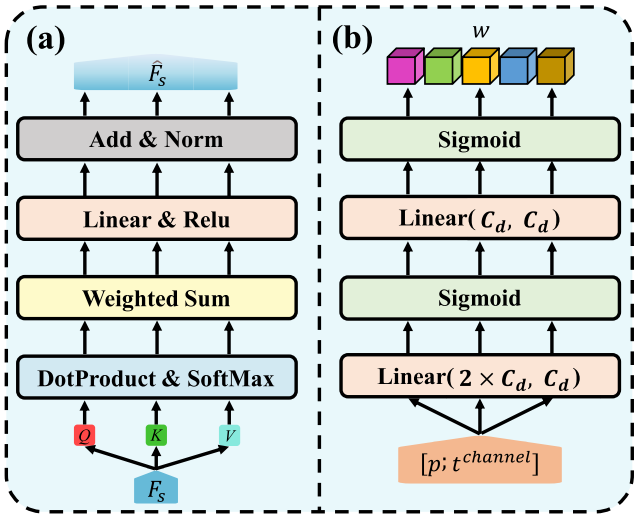


Fig. 4. Structure of (a) SDI and (b) GCM modules.

support-query image pairs with visual disparities. In contrast, individual-agnostic CD embeddings encompass general semantic knowledge. Building on this concept, we propose a SDI module to embed the semantic knowledge from the spatial dimension to supplement missing general information in visual cues.

Inspired by prompt learning methods [79], [80], we concatenate CD embeddings with visual feature maps and further promote the interaction between the two through a self-attention interactor [Fig. 4(a)], achieving semantic knowledge embedding. Given the mid-level support feature f_s of a certain category and its CD embedding $t \in \mathbb{R}^{C_t \times 1 \times 1}$, we obtain a new feature map $F_s \in \mathbb{R}^{C \times (HW+W)}$ by extending f_s with the projected CD embedding

$$F_s = \left[f_s; \text{repeat}(t^{\text{spatial}}) \right] \tag{3}$$

where $t^{\text{spatial}} = \mathcal{P}_{\text{spatial}}(t)$, $\mathcal{P}_{\text{spatial}}(\cdot)$ is a projector that maintains the channel dimension to be the same as the visual feature, $[\cdot;\cdot]$ denotes the feature concatenation operation, and repeat$(\cdot)$ expands the dimension through repetition operations.

Then, the extended feature map F_s is fed into the *interactor*, in which a self-attention block allows the flow of general semantic information to visual features. Formally, three input vectors $\boldsymbol{q}$, $\boldsymbol{k}$, and $\boldsymbol{v}$ of the attention block have the same magnitude as F_s, the output feature map $\hat{F}_s \in \mathbb{R}^{C \times H \times W}$ can be calculated by the following equation:

$$\hat{F}_s = \text{Resize}(\text{interactor}(F_s, F_s, F_s)[:, : HW]) \tag{4}$$

where Resize$(\cdot)$ restores the spatial dimension $H \times W$ of the feature map.

Finally, we further compute the general support prototype p_{gen} supplemented with the semantic knowledge as follows:

$$p_{\text{gen}} = \mathcal{A}(\hat{F}_s \otimes \mathcal{R}(M_s)). \tag{5}$$

E. Global Content Modulation

Features from channels often contain rich global semantic content, which is closely linked to the contextual understanding of the target scene. Therefore, besides embedding general semantic knowledge, i.e., CD embeddings, within the spatial dimension, it is equally crucial to modulate visual features on a channel-by-channel basis.

Given the visual support prototype $p \in \mathbb{R}^{C \times 1 \times 1 \times 1}$ of a certain category and its CD embedding $t \in \mathbb{R}^{C_t \times 1 \times 1}$, we first convert the two to an enhancement coefficient by a modulator

$$w = \text{modulator}[p; t^{\text{channel}}] \tag{6}$$

where $w \in \mathbb{R}^{C \times 1 \times 1}$, $t^{\text{channel}} = \mathcal{P}_{\text{channel}}(t)$, $\mathcal{P}_{\text{channel}}(\cdot)$ is a projector that keeps the channel dimension consistent with the visual prototype, and the structure of the *modulator* is illustrated in Fig. 4(b).

Then, the general prototype and query feature can be modulated by the enhancement coefficient

$$\begin{cases} p_{\text{gen}}^{\text{mod}} = p_{\text{gen}} \otimes w + t^{\text{channel}} \\ f_q^{\text{mod}} = f_q \otimes w + t^{\text{channel}}. \end{cases} \tag{7}$$

Through such embedding of channel-level general semantic knowledge, the global information specific to the target categories contained within the support and query branches is accentuated, and the interference caused by intraclass individual differences is effectively suppressed.

F. Prediction and Training Loss

We employ the commonly used encoder–decoder structure in segmentation tasks [34], [35]. Having obtained the modulated general prototype $p_{\text{gen}}^{\text{mod}}$ and modulated query feature f_q^{mod} from the aforementioned encoder (Sections III-C–III-E), we aggregate them into the SD $\mathcal{D}(\cdot)$ to attain the final prediction $Y \in \mathbb{R}^{H \times W}$

$$Y = \mathcal{D}\left(\left[p_{\text{gen}}^{\text{mod}}; f_q^{\text{mod}} \right] \right). \tag{8}$$

As $p_{\text{gen}}^{\text{mod}}$ incorporates a holistic embedding of general semantic knowledge across both spatial dense and global content perspectives, it is equipped to provide robust segmentation guidance for the query image. The segmentation loss is computed utilizing the standard binary cross-entropy (BCE) function

$$\mathcal{L} = \mathcal{D}\left(\left[p_{\text{gen}}^{\text{mod}}; f_q^{\text{mod}} \right] \right). \tag{9}$$

G. Extending to K-Shot Setting

In the K-shot scenario, K support pairs $\{I_s^k, M_s^k\}_{k=1}^K$ are available to provide visual segmentation guidance. Our method requires only minimal adjustments to address the K-shot setting. In the initial feature extraction stage, the K visual prototypes and query prior masks computed by (1) and (2) are averaged. In addition, K support samples spatial densely interact with their respective CD embeddings, yielding $\{p_{\text{gen}}^k\}_{k=1}^K$. The final p_{gen} is obtained by averaging these values as follows:

$$p_{\text{gen}} = \frac{1}{K} \sum_{k=1}^{K} p_{\text{gen}}^k. \tag{10}$$

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TABLE I
CATEGORY DISTRIBUTION AND SPLITTING OF iSAID-5^i DATASET

Split ID	iSAID-5^0					iSAID-5^1					iSAID-5^2				
Category ID	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15
Category name	ship	storage tank	baseball diamond	tennis court	basketball court	ground track field	bridge	large vehicle	small vehicle	helicopter	swimming pool	roundabout	soccer ball field	plane	harbor
Num of imgs in train set	3820	902	495	2485	598	1405	224	2953	2592	69	147	203	2000	1864	3358
Num of imgs in test set	1392	269	221	767	160	517	91	789	781	21	45	45	682	990	1361

IV. EXPERIMENTS

To ascertain the efficacy of the proposed HSE method, both one-shot and five-shot settings are adhered to, conducting ample quantitative and qualitative experiments on the publicly available FSS dataset. First, we delineate the FSS benchmark and the corresponding evaluation metric. Then, the implementation specifics of HSE are elucidated to facilitate realization. Following this, we present comparisons between the HSE method and other leading counterparts, analyzing the segmentation results comprehensively. Ultimately, a series of ablation studies are executed to demonstrate the impact of each component within the HSE framework.

A. Setup

1) Dataset: Despite the recent emergence of a few FSS datasets for RS images, they still lack a unified partitioning standard, making it difficult to fairly compare the performance of different algorithms. In response, we adopt the iSAID-5^i dataset [73] widely used by classical algorithms. It encompasses 15 categories, with 18 076 images designated for training and 6363 images for testing. Each sample is dimensioned at $3 \times 256 \times 256$. The specific distribution of categories and their partitioning are illustrated in Table I.

2) Evaluation Metric: Building upon prior research in the domain [73], [75], we utilize the mean intersection over union (mIoU) as the evaluation metric. Formally, let N be the number of test (unseen) categories in each target fold, and the IoU of category c is represented by IoU_c, the mIoU can be defined as follows:

$$\text{mIoU} = \frac{1}{N}\sum_{c=1}^{N}\text{IoU}_c. \tag{11}$$

B. Implementation Details

HSE is constructed using the PyTorch [86] framework, and all experiments are conducted on a single NVIDIA GeForce RTX 3090 GPU. We leverage ResNet50 [44] and VGG16 [45] pretrained on ImageNet [87] as feature extractors. The baseline process involves removing the SDI and GCM modules from the proposed HSE, which directly combines the support prototype, query feature, and query prior mask [26], and then feeding them into the SD. To preserve the generalizability of the feature extractor, we follow [26] to freeze its parameters. To derive general semantic knowledge from class name embeddings, three types of language models pretrained on large-scale corpora are utilized, i.e., CLIP [43], SBERT [88], and GloVe [89]. We augment the category name with a text template as input: *A photograph of a "category name."*

During meta-training, we employ the stochastic gradient descent (SGD) optimizer with an initial learning rate of 0.005, a momentum of 0.9, and a weight decay of 0.0001. The batch size is set at 16 and the training epoch is fixed at 100.

During the meta-testing phase, we randomly sample 2000 episodes for model evaluation, with the final results obtained by averaging the outcomes of experiments influenced by five different random seeds.

C. Comparison With State-of-the-Art

In this section, we compare the proposed HSE method against other advanced FSS algorithms with the available source codes or public experimental results on the iSAID-5^i dataset. Under standardized experimental conditions, the quantitative and qualitative comparison results are presented to rigorously validate the superiority of HSE.

1) Quantitative Analysis: The performance comparison of our proposed HSE against other exemplary FSS algorithms is depicted in Table II, and mIoU is leveraged as the evaluation metric under one-shot and five-shot settings. The best performance values are highlighted in bold, and the second-best results are marked with an underline. Overall, the proposed HSE method achieves the best segmentation performance across various backbones and support sample sizes. Specifically, with the ResNet50 backbone, HSE achieves the best performance in every split, enhancing the average mIoU by 2.73% under one-shot settings. For the five-shot setting, HSE exhibits a distinct advantage, achieving an average performance of 43.15%, representing an improvement of 2.09%. Simultaneously, employing the VGG16 backbone, HSE achieved the best performance for each split under the one-shot setting, improving the average performance by 0.97%. Like the five-shot setting results, HSE led in both split1 and split2, surpassing the HMRE method by an average mIoU of 1.51%. In terms of comprehensive analysis, the proposed HSE method demonstrates significant advantages in FSS tasks, particularly excelling in one-shot scenarios. This effectively validates that incorporating general semantic knowledge is beneficial for establishing robust class-specific segmentation guidance.

To specifically analyze the model's performance across various categories, we detail the segmentation results under the one-shot setting in Table III. The proposed HSE achieves excellent performance across the majority of categories. While some methods get better segmentation results in specific

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TABLE II

PERFORMANCE COMPARISON ON iSAID-5^i DATASET UNDER ONE-SHOT AND FIVE-SHOT SETTINGS. BOLDED VALUES INDICATE THE BEST PERFORMANCE, WHILE THE SECOND-BEST PERFORMANCES ARE UNDERLINED

Method	1-shot				5-shot			
	Split0	Split1	Split2	Mean	Split0	Split1	Split2	Mean
ResNet50								
PANet [21]	27.56	17.23	24.60	23.13	36.54	16.05	26.22	26.27
CANet [22]	25.51	13.50	24.45	21.15	29.32	21.85	26.91	26.03
SCL [38]	34.78	22.77	31.20	29.58	41.29	25.73	37.70	34.91
PFENet [26]	35.84	23.35	27.20	28.80	42.42	25.34	33.00	33.59
NERTNet [81]	34.93	23.95	28.56	29.15	44.83	26.73	37.19	36.25
DCP [27]	37.83	22.86	28.92	29.87	41.52	28.18	33.43	34.38
BAM [82]	39.43	21.69	28.64	29.92	43.29	27.92	38.62	36.61
MIANet [23]	41.43	27.26	35.70	34.80	45.75	32.34	45.21	41.10
DMML [74]	28.45	21.02	23.46	24.31	30.61	23.85	24.08	26.18
DML [83]	32.96	18.98	26.27	26.07	33.58	22.05	29.77	28.47
SDM [33]	27.96	21.99	27.82	25.92	28.50	25.23	31.07	28.27
SD-AANet [84]	30.52	29.31	18.46	26.10	28.01	21.38	16.79	22.06
TBPN [85]	29.33	16.84	25.47	23.88	30.98	20.42	28.07	26.49
HMRE [68]	40.88	32.88	37.22	36.99	42.41	34.69	**46.07**	41.06
R^2Net [77]	41.22	21.64	35.28	32.71	46.45	25.80	39.84	37.36
PCNet [75]	40.24	24.64	31.31	32.06	45.31	28.19	37.36	36.95
HSE(*ours*)	**44.03**	**35.87**	**39.26**	**39.72**	**47.03**	**36.75**	45.66	**43.15**
VGG16								
PANet [21]	26.86	14.56	20.69	20.70	30.89	16.63	24.05	23.86
CANet [22]	13.91	12.94	13.67	13.51	17.32	15.07	18.23	16.87
SCL [38]	25.75	18.57	22.24	22.19	35.77	24.92	32.70	31.13
PFENet [26]	28.52	17.05	18.94	21.50	37.59	23.22	30.45	30.42
NERTNet [81]	25.78	20.01	19.88	21.89	38.43	24.21	28.99	30.54
DCP [27]	28.17	16.52	22.49	22.39	39.65	22.68	29.93	30.75
BAM [82]	33.93	16.88	21.47	24.09	38.46	22.76	28.81	30.01
MIANet [23]	35.56	25.89	34.17	31.87	41.33	24.72	40.42	35.49
DMML [74]	24.41	18.58	19.46	20.82	28.97	21.02	22.78	24.26
DML [83]	30.99	14.60	19.05	21.55	34.03	16.38	26.32	25.58
SDM [33]	24.52	16.31	21.01	20.61	26.73	19.97	26.10	24.27
SD-AANet [84]	28.01	21.38	16.79	22.06	34.89	25.76	16.81	25.82
TBPN [85]	27.86	12.32	18.16	19.45	32.79	16.28	24.27	24.45
HMRE [68]	36.48	28.28	35.36	33.37	37.33	29.28	43.36	36.66
R^2Net [77]	35.27	19.93	24.63	26.61	**42.06**	23.52	30.06	31.88
PCNet [75]	32.48	19.88	24.56	25.64	41.09	21.98	34.14	32.40
HSE(*ours*)	**37.18**	**29.79**	**36.04**	**34.34**	40.87	**30.20**	**43.44**	**38.17**

categories, their performance is not consistently stable. For example, the BAM algorithm performs well in segmenting tennis court (C4), but it struggles with smaller sized targets such as harbor (C15). The DML method also excels at parsing tennis court (C4), yet it encounters difficulties with the small vehicle category (C9). In contrast, our method still maintains good segmentation capabilities for challenging categories, exhibiting a more stable overall performance. This indicates that HSE has a stronger adaptability to complex scenarios, benefiting from the general CD embedding that compensates for the lack of sparse visual representation information.

2) Qualitative Analysis: To visually analyze the efficacy of the proposed HSE, Fig. 5 illustrates quantitative segmentation results in the one-shot scenario, where the baseline approach represents HSE with the GCM and SDI components removed. Compared with the baseline method, HSE accurately activates target scene areas, generating prediction masks with more finely detailed edges. Moreover, HSE effectively reduces false activations in query image areas similar to the target category. Particularly for the "ship" and "swimming pool" categories, the baseline method misclassifies similar class regions in the background, while HSE effectively suppresses the aforementioned interference.

We visualize the impact of introducing CD embeddings on the attention map in SDI. The second row in Fig. 6 represents self-attention applied solely to visual features f_s

$$\hat{f}_s = \text{interactor}(f_s, f_s, f_s). \tag{12}$$

The second row shows the attention map $\hat{F}_s$ with the introduction of CD embeddings, as in (4). The introduction of CD embeddings enhances the visual information shortfall by steering the model's focus toward spatial features specific to each class, thus providing more reliable segmentation guidance.

Additionally, Fig. 7 illustrates the t-SNE [90] visualization results under the five-way one-shot 20-query setting. The baseline method is detailed in Section IV-B. It is evident that the proposed method significantly improves intraclass consistency. Furthermore, introducing the CD embedding enhances the distribution of the support representation (segmentation guidance), allowing it to better cover intraclass differences.

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TABLE III

PERFORMANCE COMPARISON ACROSS SPECIFIC CATEGORIES ON iSAID-5^i DATASET UNDER THE ONE-SHOT SETTING

Category ID	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15
ResNet50															
PANet [21]	21.81	36.31	23.01	42.06	14.59	12.11	17.44	22.70	12.27	21.60	30.29	24.62	26.79	25.54	15.79
CANet [22]	39.57	18.54	18.46	33.63	17.34	9.78	5.49	22.15	5.17	24.89	9.96	36.50	19.12	38.82	17.85
SCL [38]	37.61	33.63	26.68	54.75	21.22	22.60	24.40	30.22	6.71	29.93	**33.00**	44.68	18.25	**44.63**	15.46
PFENet [26]	39.02	45.63	20.86	49.96	23.72	21.00	24.76	31.59	6.98	32.42	13.34	47.64	30.65	32.82	11.54
NERTNet [81]	33.59	42.83	22.30	49.35	21.91	21.62	28.82	25.64	9.35	34.30	23.91	38.67	25.63	40.84	13.74
DCP [27]	37.42	42.44	35.16	56.55	17.58	21.66	19.57	32.97	10.60	29.50	24.02	35.34	28.44	39.80	17.02
BAM [82]	36.34	39.76	38.23	**58.13**	24.71	18.25	12.68	35.91	11.42	30.21	28.98	40.74	29.43	33.25	10.79
DMML [74]	40.14	40.18	21.31	27.02	13.60	15.56	15.19	26.05	13.84	**34.44**	11.26	17.57	23.27	39.11	26.12
DML [83]	35.13	42.10	30.49	41.79	15.31	13.25	16.87	24.70	14.62	25.45	10.24	35.49	25.35	41.69	18.57
SDM [33]	41.77	35.50	21.41	20.81	20.29	15.60	25.60	28.66	13.29	26.79	13.61	32.35	24.59	42.79	25.75
SD-AANet [84]	25.21	39.85	**43.60**	21.69	22.27	**35.18**	**47.28**	23.89	25.47	14.75	9.46	27.38	20.51	12.50	22.45
TBPN [85]	25.36	41.28	30.67	32.88	16.48	13.48	9.74	27.88	12.52	20.56	11.12	34.31	23.57	40.36	17.98
HMRE [68]	41.68	41.15	40.01	41.04	40.52	27.36	34.66	35.44	**39.34**	27.58	24.56	49.34	**46.82**	4.27	21.11
R^2Net [77]	**46.87**	**49.06**	30.70	52.86	26.62	24.31	17.25	31.25	13.67	21.73	24.88	46.07	42.29	42.07	21.08
HSE(*ours*)	45.67	46.21	43.34	41.02	**43.91**	33.40	40.08	**36.13**	39.30	30.44	28.58	**50.61**	45.89	43.50	**27.72**
VGG16															
PANet [21]	20.05	37.71	21.18	41.22	14.15	12.17	13.82	21.05	7.89	17.88	4.36	31.68	27.55	26.88	12.97
CANet [22]	24.13	6.73	13.83	16.32	8.54	14.12	3.24	21.04	3.35	22.96	9.57	14.91	17.83	16.11	9.92
SCL [38]	28.50	32.93	19.68	29.60	18.05	22.48	7.92	31.46	8.99	22.02	14.17	16.53	19.72	**39.40**	21.37
PFENet [26]	34.32	31.81	24.20	35.43	16.86	13.98	6.01	31.68	6.76	**26.85**	8.15	17.75	20.56	33.34	14.87
NERTNet [81]	12.66	23.11	26.90	50.47	15.77	23.14	8.48	31.73	11.75	24.94	14.64	20.45	29.03	28.06	7.24
DCP [27]	27.69	38.45	25.92	33.20	15.57	17.62	12.36	26.79	8.05	17.80	22.45	18.29	18.03	37.57	16.10
BAM [82]	27.66	43.90	31.48	43.96	22.66	13.57	8.91	31.76	9.26	20.91	17.05	26.27	30.68	25.27	8.07
DMML [74]	34.75	37.36	15.15	22.85	11.94	21.41	13.85	23.92	10.24	23.50	8.17	16.32	21.08	29.63	22.09
DML [83]	27.30	42.63	19.25	**50.63**	15.13	14.16	15.94	22.40	7.74	12.74	3.79	23.73	23.47	27.40	16.88
SDM [33]	33.76	23.88	17.80	27.76	19.38	18.36	9.63	25.24	8.63	19.69	10.56	15.36	24.76	32.30	22.06
SD-AANet [84]	18.72	32.01	40.17	23.59	25.58	13.79	**48.74**	13.67	14.93	15.79	8.36	22.03	18.86	11.05	23.66
TBPN [85]	22.03	39.75	20.80	42.80	13.94	10.41	6.87	16.54	4.38	23.41	5.68	23.66	22.13	24.63	14.72
HMRE [68]	30.25	36.16	**41.88**	43.37	30.74	31.52	36.66	**35.10**	18.38	19.74	9.74	**45.09**	**49.22**	30.82	21.93
R^2Net [77]	**37.82**	**45.16**	26.27	45.30	21.81	24.11	14.38	30.92	12.21	18.03	18.66	25.02	29.64	31.95	17.87
HSE(*ours*)	37.56	40.21	40.65	36.20	**31.28**	**34.30**	38.24	33.26	**20.01**	23.14	**32.70**	43.45	47.63	30.89	**25.53**

TABLE IV

ABLATION STUDY FOR EACH COMPONENT OF HSE UNDER THE ONE-SHOT SETTING, WITH THE BEST RESULTS HIGHLIGHTED IN BOLD

1-Shot Setting, with the Best Results Highlighted in Bold.

Methods	1-shot				Params	Inference time
	Split0	Split1	Split2	Mean		
Baseline	38.25	30.88	35.06	34.73	26.3M	34.78ms
Baseline+SDI	42.55	34.2	37.98	38.24	26.7M	35.25ms
Baseline+GSM	41.01	33.17	36.66	36.95	26.6M	35.68ms
Baseline+GSM+SDI	**44.03**	**35.87**	**39.26**	**39.72**	27.0M	36.15ms

TABLE V

IMPACT OF DIFFERENT LANGUAGE MODELS ON THE QUALITY OF GENERAL SEMANTIC KNOWLEDGE UNDER THE ONE-SHOT SETTING

Language model	1-shot			
	Split0	Split1	Split2	Mean
SBERT	43.58	35.26	38.51	39.12
GloVe	43.65	**35.90**	39.14	39.56
CLIP	**44.03**	35.87	**39.26**	**39.72**

D. Ablation Studies

1) Component Analysis: To verify the effectiveness of each module within the proposed HSE framework, we organize component-level ablation experiments. As shown in Table IV, the SDI and GCM modules contribute significantly to performance, each enhancing mean mIoU by 3.51% and 1.55%, respectively. This demonstrates that embedding general semantic knowledge from both spatial dense and global content perspectives is beneficial for establishing superior segmentation guidance. More importantly, the two modules complement each other, synergizing to enhance the performance of the proposed HSE further.

Regarding resource overhead, since the CLIP text encoder is only involved in a one-time computation, its parameters (151 M) and time consumption (0.839 s) are excluded from the overall evaluation. It can be observed that our designed SDI and GCM modules are highly efficient in computation while introducing only a minimal increase in model parameters.

2) Language Model Selection: To explore the impact of general semantic knowledge produced by different language models on segmentation performance, Table V displays the results using pretrained SBERT, GloVe, and CLIP models, respectively, in the one-shot scenario. It is evident that HSE effectively accommodates different language models and consistently achieves excellent segmentation performance. Overall, CLIP holds a slight advantage, likely because its pretraining regime better aligns visual and linguistic concepts. We select CLIP as the default language model in other experiments.

3) Projector Structure: In the design of the SDI and GCM modules, CD embeddings t are mapped to t^{channel} and t^{spatial} using $\mathcal{P}_{\text{channel}}(\cdot)$ and $\mathcal{P}_{\text{spatial}}(\cdot)$, respectively. To identify the

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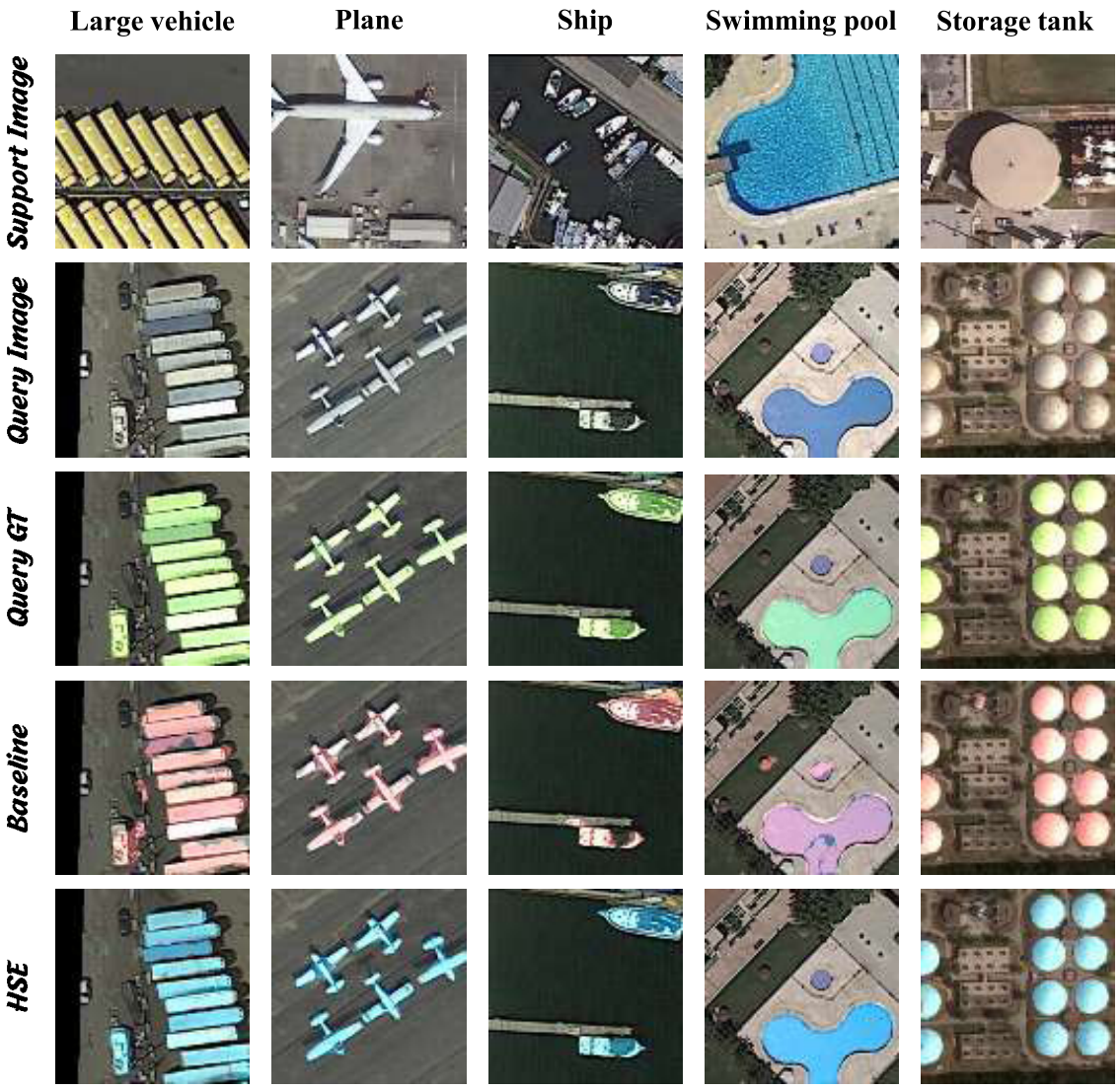


Fig. 5. Quantitative comparison of the segmentation effects of the HSE and baseline methods on iSAID-5^i dataset under the one-shot setting.

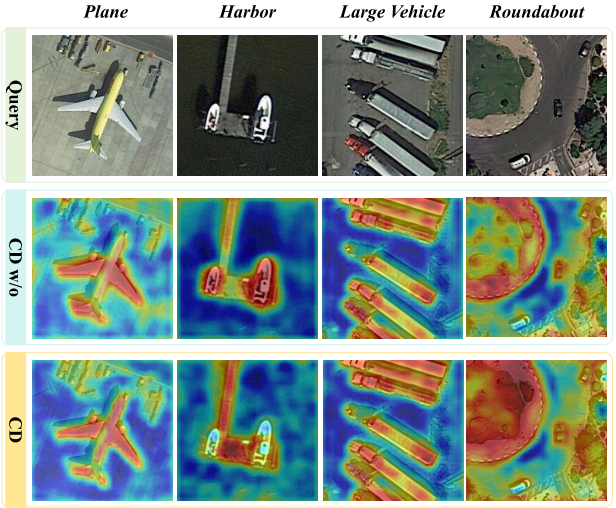


Fig. 6. Visualization of the impact of introducing CD embeddings on the attention map.

optimal projector, comparative experiments are conducted with configurations of *Linear*, *two-layers-MLP*, and *three-layers-MLP*. From Table VI, it can be observed that the structure of the projector has a minimal impact on the model's segmentation performance. Although the *two-layers-MLP* exhibits a slight advantage, considering the overhead of model parameters, we choose the *Linear* as the default projector.

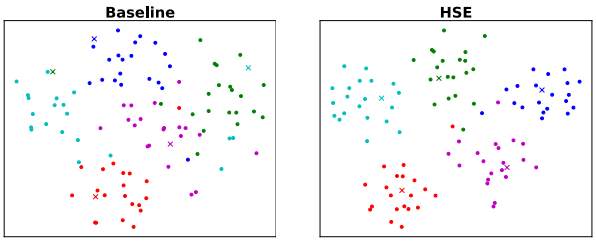


Fig. 7. t-SNE visualization of the support and query features. • represents the query features of different categories, while × marks indicate the corresponding support features for each category.

4) Various Textual Descriptions: Table VII presents the impact of embedding different textual descriptions on model performance under the one-shot setting. Unlike natural images,

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TABLE VI
IMPACT OF DIFFERENT PROJECTOR STRUCTURES ON SEGMENTATION PERFORMANCE UNDER THE ONE-SHOT SETTING

Projector	1-shot			
	Split0	Split1	Split2	Mean
Linear	44.03	**35.87**	39.26	39.72
2-layers-MLP	**44.15**	35.73	**39.45**	**39.78**
3-layers-MLP	43.34	34.26	37.52	38.37

TABLE VII
IMPACT OF DIFFERENT TEXT DESCRIPTIONS ON SEGMENTATION PERFORMANCE UNDER THE ONE-SHOT SETTING

1-Shot Setting.

Text descriptions	1-shot			
	Split0	Split1	Split2	Mean
A photo of a {*category*}	43.52	35.82	38.78	39.37
A scene of a {*category*}	43.89	35.25	38.61	39.25
There is a {*category*} in the image	43.87	35.66	**39.40**	39.64
{*category*}	**44.03**	**35.87**	39.26	**39.72**

where detailed template descriptions generally enhance performance, we observed that using simple category names yields better results in RS imagery. We attribute this to the inherent complexity of RS scenes, which encompass diverse object elements. Describing a specific category as a scene or image tends to be inaccurate, thereby diminishing the effectiveness of textual knowledge embedding.

5) Design of the SDI and GCM Modules: We investigate the impact of various implementation strategies for the SDI and GCM modules on segmentation performance. We evaluate three SDI strategies (sd1, sd2, sd3, and sd4) and two GCM strategies (gc1 and gc2). Specifically, sd1 represents the addition of the mid-level support feature f_s and the projected CD embedding t^{spatial}, sd2 denotes the element-wise multiplication of f_s and t^{spatial}, sd3 stands for concatenating the visual prototype p and t^{spatial} along the channel dimension, followed by two fully connected layers as in MIANet [23], and sd4 involves concatenating f_s and t^{spatial} followed by processing with self-attention, as formulated in Section III-D. gc1 represents modulating the general prototype p_{gen} and the mid-level query feature f_q using only the enhancement coefficient w, while gc2 employs a residual modulation approach, as described in Section III-E. Fig. 8 displays the results under different combinations of implementation strategies, where the optimal performance is obtained with the combination of sd4 and gc2. This demonstrates the superiority of the methods proposed in this article.

6) Number of Support Samples: In Fig. 9, with the ResNet50 backbone, we illustrate the impact of varying numbers of support samples on the segmentation results for the one-shot task. As the number of support samples increases from 1 to 9, incorporating richer visual representations leads to a gradual improvement in performance. However, when the number of support samples exceeds five, the incremental improvement in segmentation performance becomes marginal. This may be due to the redundancy of target-category-relevant information in the numerous support samples. Additionally, as visual representations dominate the segmentation guidance, the benefits brought by the introduced generalized semantic knowledge are gradually diminished.

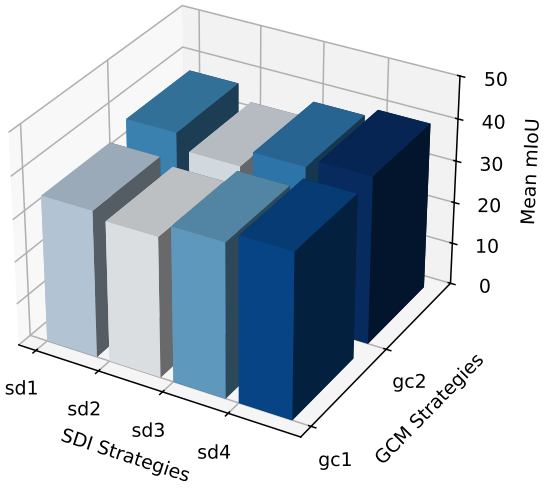


Fig. 8. Ablation studies on the different designs of SDI and GCM modules under the one-shot setting.

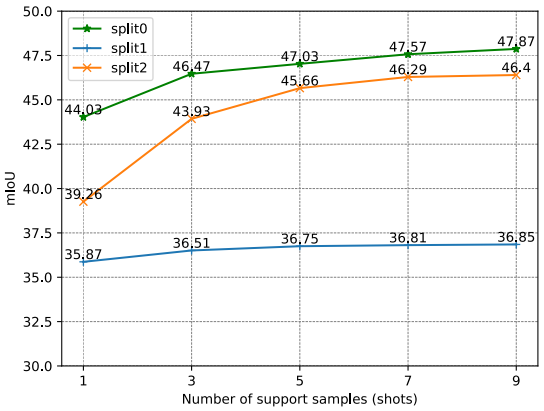


Fig. 9. Impact of the number of support samples on segmentation performance under the one-shot setting.

V. CONCLUSION

In this article, we proposed the HSE method for the FSS task in RS imagery. Addressing the particularly severe issue of intraclass differences in RS images, we highlight that relying solely on sparse visual support representations is insufficient to establish robust segmentation guidance. Based on this observation, as the first attempt to introduce general semantic knowledge from CD embeddings into this field, we designed two complementary and sequential modules: SDI and GCM. They respectively supplement the missing target scene spatial information in sparse visual representations and modulate the global content related to target categories. Experiments conducted on the iSAID-5^i dataset validate the effectiveness of the designs proposed in this article, while also demonstrating that the introduced HSE method sets a new benchmark for excellence.

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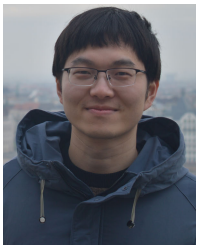
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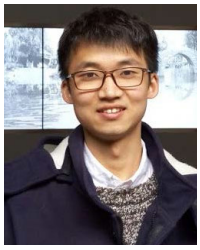
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